

Seat No.: _____

Enrolment No. _____

NATIONAL FORENSIC SCIENCES UNIVERSITY

M.Sc. Cyber Security- Semester End Examination-July-2023

Subject Code: CTMSCS SII P2

Subject Name: Malware Analysis

Time: 11:00 AM to 2:00 PM

Date: /07/2023

Total Marks: 100

023

00

Instructions:

1. Write down each question on separate page.
2. Attempt all questions.
3. Make suitable assumptions wherever necessary.
4. Figures to the right indicate full marks.

			Marks
Q.1	(a)	How to detect the packer? Explain any three methods.	04
	(b)	How to analyzed dynamic linking and run-time linking libraries.	04
	(c)	Explain the Unpacker stub and unpacking methods in detail. OR Set-up an ideal lab for dynamic analysis of the malware.	08
	(d)	Illustrate the importance of malware analysis in the industrial aspects.	09
Q.2	(a)	Explain PE Header and its sections. OR Discuss pros and cons of Sandbox.	04
	(b)	Discuss Disassembly and their types/algorithms.	04
	(c)	Explain CPU Register their types along with the examples.	08
	(d)	Explain at least 4 types of malwares with their characteristics, behaviour and real life case study in brief.	09
Q.3	(a)	Three functions are commonly found in cases of direct injection: VirtualAllocEx , WriteProcessMemory , and CreateRemoteThread. Explain them briefly.	04
	(b)	Explain the key-logger behavior from Figure 2.0 from page no. 2 OR State the difference between Kernal mode and User mode debugging.	04
	(c)	Explain Reverse Engineering process and some important features of IDA Pro.	08
	(d)	Write a detailed note on Breakpoint and its types.	09
Q.4	(a)	What is Hook Injection?	04
	(b)	Explain four major section of main memory.	04
	(c)	Bob is analyzing a malware and he finds an interesting code snippets while doing reverse engineering. Bob needs help from you to understand the code snippet shown at figure 1.0 in page no.2	08

	(d) Discuss Anti-Reverse Engineering techniques with its types and examples. OR Explain any 9 assembly instructions with proper example and description.	09
--	---	----

```

0040286F  push    2                ; samDesired
00402871  push    eax               ; ulOptions
00402872  push    offset SubKey     ; "Software\\Microsoft\\Windows\\CurrentVersion\\Run"
00402877  push    HKEY_LOCAL_MACHINE ; hKey
0040287C  call    esi               ; RegOpenKeyExW
0040287E  test    eax, eax
00402880  jnz     short loc_4028C5
00402882
00402882  loc_402882:
00402882  lea     ecx, [esp+424h+Data]
00402886  push    ecx               ; lpString
00402887  mov     bl, 1
00402889  call    ds:strlenW
0040288F  lea     edx, [eax+eax+2]
00402893  push    edx               ; cbData
00402894  mov     edx, [esp+428h+hKey]
00402898  lea     eax, [esp+428h+Data]
0040289C  push    eax               ; lpData
0040289D  push    1                 ; dwType
0040289F  push    0                 ; Reserved
004028A1  lea     ecx, [esp+434h+ValueName]
004028A8  push    ecx               ; lpValueName
004028A9  push    edx               ; hKey
004028AA  call    ds:RegSetValueExW

```

Figure – 1.0

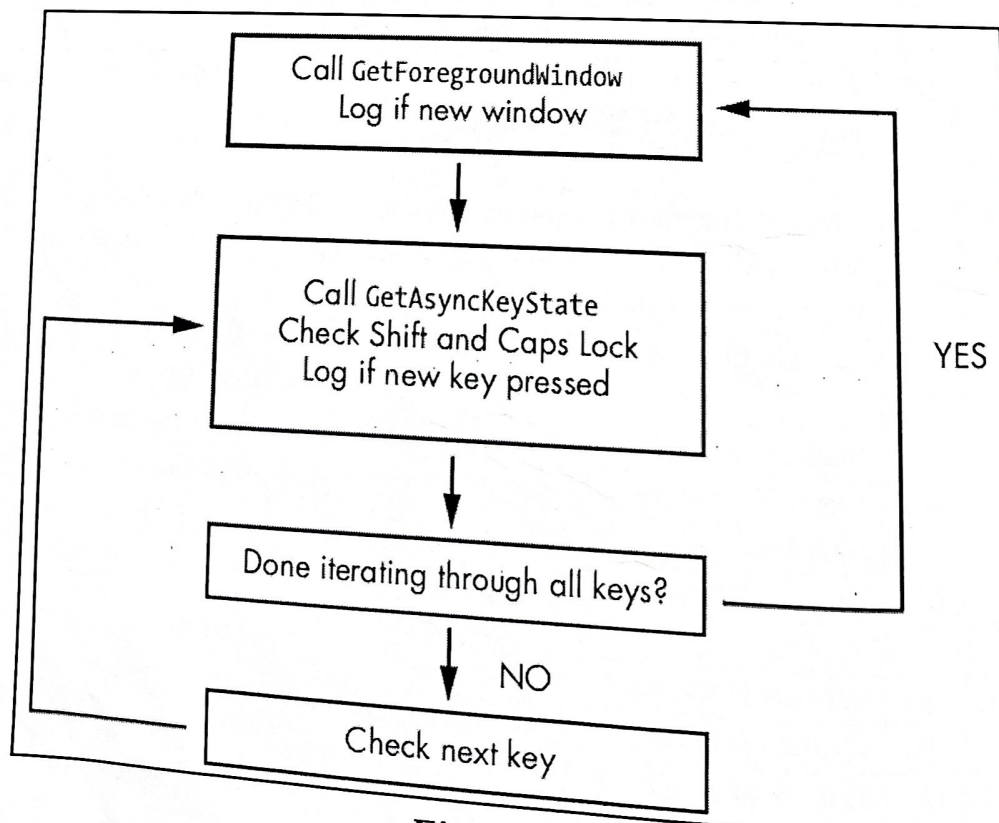


Figure - 2.0

Seat No.: _____

Enrolment No. _____

NATIONAL FORENSIC SCIENCES UNIVERSITY

M.Sc. Cyber Security – ATKT - Examination-Nov-2023

Subject Code: CTMSCS SII P2
Subject Name: Malware Analysis
Time: 11:00 AM to 2:00 PM

Date: 29/11/2023

Total Marks: 100

Instructions:

1. Write down each question on separate page.
2. Attempt all questions.
3. Make suitable assumptions wherever necessary.
4. Figures to the right indicate full marks.

			Marks
Q.1	(a)	What do you mean by malware? Explain it in brief.	04
	(b)	Write a short note on Trojan type of malware.	04
	(c)	What are the limitations of static analysis techniques? How it can be covered? Explain the solution with its advantage in detail.	08
	(c)	OR Explain the PE header and its sections in detail with respect to the malware analysis.	08
	(d)	There is an attack in the organization. The computers and servers are encrypted and asked to pay the ransom in order to get the decryption key. You have been given the sample of that malware and you are the malware analyst. Explain the process and steps that you will follow to analyze that malware.	09
Q.2	(a)	Explain Anti-Virus Signature with an example.	04
	(b)	How to use ClamAV and PEView tool?	04
	(c)	Discuss the packers and obfuscation in detail.	08
	(d)	OR Explain Anti-Reverse Engineering Techniques in detail.	09
	(d)	Discuss any real-world case study of well known malware with its attack vector, kill chain and technical details.	09
Q.3	(a)	Define any four DLLs.	04
	(b)	Briefly explain the android architecture.	04
	(b)	OR State the difference between user mode and kernel mode.	04
	(c)	Explain the reverse engineering assembly of the following program: <i>i = 2;</i> <i>j = user defined value;</i> <i>if (j % i == 0)</i>	08

		<pre> { printf("the number is even"); } else { printf("the number is odd"); } </pre>	
	(d)	Write a detailed note on Process injection and Process replacement.	09
Q.4	(a)	How to analyze dll (linked libraries) from the point of view of malware analysis.	04
		OR	
	(a)	Explain various Flag Registers with assembly example.	04
	(b)	Explain Windows Registry in brief and also discuss the working of REGSHOT.	04
	(c)	Illustrate the use of Debugger for malware analysis in detail.	08
	(d)	Discuss the importance of IDA pro with its features with respect to malware analysis in detail.	09

NATIONAL FORENSIC SCIENCES UNIVERSITY**M.Sc. Cyber Security - Semester – II – (ATKT Exam) - Dec-2023****Subject Code: CTMSCS SII P3****Date: 01/12/2023****Subject Name: Mobile Security****Time: 11:00 AM to 2:00 PM****Total Marks: 100****Instructions:**

1. Write down each question on a separate page.
2. Figures to the right indicate full marks.

	Marks
Q.1 Attempt Any 3 Questions (8 Marks each)	24
(a) What is Intent? Explain types of Intent with example.	
(b) What is the use of Android Manifest? What we can write into Manifest file. Explain it with example of Manifest file.	
(c) Explain Android Architecture with all layers?	
(d) What is Application Sandboxing? Explain Secure IPC.	
Q.2 Attempt Any 3 Questions (8 Marks each)	24
(a) What is Content Provider? Explain with example of writing the log.	
(b) Explain Android Boot Process in Detail.	
(c) What is ADB? Explain ADB Commands.	
(d) Explain Insecure Logging and Hardcoding Issues?	
Q.3 Attempt Any 3 Questions (8 Marks each)	24
(a) What is Reverse Engineering? Write a Steps to extract source code from APK.	
(b) What is the use of JADX, JDGUI and DexDump Reverse Engineering	
(c) Differentiate Static Analysis Vs Dynamic Analysis	
(d) Explain Mobile Application Security Pen-Testing Strategy	
Q.4 Attempt Any 2 Questions (7 Marks each)	14
(a) Explain any 7 Commands of Drozer with syntax and example.	
(b) Explain Dynamic Instrumentation using Frida with commands.	
(c) Explain QARK with features, advantages, modes, pros and cons.	
Q.5 Attempt Any 2 Questions (7 Marks each)	14
(a) Explain Step By Step Active Analysis of Android Traffic using Burp Suite.	
(b) Explain HTTPS Proxy interception with example.	
(c) What is Android Traffic Interception. Explain ways to analyze the Android traffic.	

END OF PAPER

Seat No.: _____

Enrolment No. _____

NATIONAL FORENSIC SCIENCES UNIVERSITY

M.Sc. Cyber Security – ATKT - Semester - II - Apr-2024

Subject Code: CTMSCS SII P2
Subject Name: Malware Analysis
Time: 2:00 PM to 05:00 PM

Date: 16-04-2024

Total Marks: 100

Instructions:

1. Write down each question on separate page.
2. Attempt all questions.
3. Make suitable assumptions wherever necessary.
4. Figures to the right indicate full marks.

- | | | Marks |
|------------|--|--------------|
| Q.1 | (a) What is malware? Discuss briefly. | 04 |
| | (b) Discuss the basic dynamic analysis in brief. | 04 |
| | (c) Discuss IDA Pro and its features. | 08 |
| | OR | |
| | (c) Write a note on Breakpoint and its types. | 08 |
| | (d) Write a detailed note on reverse engineering process of malware. | 09 |
| Q.2 | (a) State the difference between debugging and disassembly. | 04 |
| | (b) Give example of "if statement" and show its assembly. | 04 |
| | (c) Discuss various malware analysis techniques in detail. | 08 |
| | (d) Explain the volatile memory analysis with volatility command. | 09 |
| | OR | |
| | (d) Write a detailed note on anti -reverse engineering. | 09 |
| Q.3 | (a) What is kernel mode debugging? | 04 |
| | OR | |
| | (a) Discuss DLL injection. | 04 |
| | (b) Explain the below mentioned reverse engineering program: | 04 |
| | <pre>00401036 mov [ebp+var_4], 0 0040103D mov [ebp+var_8], 0 00401044 loc_401044: 00401044 cmp [ebp+var_4], 0 00401048 jnz short loc_401063 ① 0040104A call performAction 0040104F mov [ebp+var_8], eax 00401052 mov eax, [ebp+var_8] 00401055 push eax 00401056 call checkResult 0040105B add esp, 4 0040105E mov [ebp+var_4], eax 00401061 jmp short loc_401044 ②</pre> | |
| | (c) Write a detailed note on "PE Header and its sections". | 08 |
| | (d) Draw and explain the android architecture. | 09 |

- Q.4** (a) What is Process injection? 04
(b) Discuss important features of ollydbg. 04

OR

- (b) Explain malware signature with one example. 04
(c) Write a note on CPU registers. 08
(d) When the developer wants to build an application for windows operating system they need to use some inbuilt OS functions / Windows API. Answer the following question. 09
a. How those function can be used in that application with different ways?
b. While analyzing this application with static and dynamic methods, how can you get the details about that inbuilt used OS functions / Windows APIs?
Explain the process with required tools.

Seat No.:

Enrolment No.:

NATIONAL FORENSIC SCIENCES UNIVERSITY

M.Sc. Cyber Security
Semester – II – September- 2024
FINAL EXAM - REMEDIAL

Subject Code: CTMSCS SII PI
Subject Name: Network Security
Time: 2:00 PM to 5:00 PM

Date: 25/09/2024

Total Marks: 100

Instructions:

1. Write down each question on a separate page.
2. Attempt all questions.
3. Make suitable assumptions wherever necessary.
4. Figures to the right indicate full marks.

Q.1	Attempt any three.	Marks
(a)	Explain following devices Networking devices : 1. Hub 2. Switch 3. Bridge 4. Router	08
(b)	Explain following Perimeter devices a) IDS b) IPS	08
(c)	Write short notes on SIEM.	08
(d)	Explain following servers 1. DNS 2. DHCP 3. Proxy 4. Mail	08
Q.2	Attempt any three.	
(a)	Write short notes on the Penetration Life cycle.	08
(b)	Explain Dumpster diving and war driving.	08
(c)	Discuss Active scanning and Passive scanning	08
(d)	Difference between UDP and TCP.	08
Q.3	Attempt any three.	
(a)	Explain SNMP service enumeration.	08
(b)	Routing devices enumeration and countermeasure.	08
(c)	Explain different password cracking tools and methods.	08
(d)	Explain digital signature working.	08
Q.4	Attempt any two.	
(a)	Describe the four-way handshake process used in WPA2.	07
(b)	Explain 802.11 protocols.	07
(c)	Explain WLAN based attack..	07
Q.5	Attempt any two.	
(a)	Write a short note on Network based digital evidence.	07
(b)	Explain Network Forensic investigation method.	07

(c) Write a short note on sources of network-based evidence.

--- End of Paper---

Serial No. _____

Enrolment No. _____

NATIONAL FORENSIC SCIENCES UNIVERSITY

M.Sc. Cyber Security - Semester - II - Sept-2024

Subject Code: CTMSCS SH P2
Subject Name: Malware Analysis
Time: 2:00 PM TO 5:00 PM
Total Marks: 100

Date: 26/09/2024

Instructions:

1. Write down each question on separate page.
2. Attempt all questions.
3. Make suitable assumptions wherever necessary.
4. Figures to the right indicate full marks.

		Marks
Q.1	Attempt any three.	
	(a) Explain Header Analysis of PE Files.	08
	(b) Write a note on Process Injection.	08
	(c) Write a note on Stuxnet.	08
	(d) What is Register? Explain any 4 computer Register.	08
Q.2	Attempt any three.	
	(a) Explain Android Architecture.	08
	(b) What is Sandbox? Explain any one sandbox in detail.	08
	(c) Explain any 3 Sysinternal Tool.	08
	(d) Explain ollyDbg in Detail.	08
Q.3	Attempt any three.	
	(a) Explain Malware Analysis with dynamic analysis technique.	08
	(b) Explain 4 malware types in details with Example.	08
	(c) Explain the Permission on Android Application.	08
	(d) What is Register? Explain any 4 computer Register.	08
Q.4	Attempt All.	
	(a) Explain Static analysis of ELF file.	07
	(b) Explain the component of Android.	07
Q.5	Attempt All.	
	(a) What is DLL? Explain any 3 DLL in detail.	07
	(b) Discuss the ELF File Types.	07

Seat No.:

Enrolment No.:

NATIONAL FORENSIC SCIENCES UNIVERSITY

M.Sc. Cyber Security
Semester – II – July - 2024

Subject Code: CTMSCS SII P1
Subject Name: Network Security
Time: 2:30 PM to 5:30 PM

Date: 10/07/2024

Total Marks: 100

Instructions:

1. Write down each question on a separate page.

Q.1	Attempt any three.	Marks
(a)	Explain OSI Model.	08
(b)	Explain following terms : 1) DNS 2) DHCP 3) Proxy 4) Application Server	08
(c)	Explain following terms : 1) Threat 2) vulnerability 3) attack surface 4) attack vector	08
(d)	Discuss application of following devices with its full form: 1) IDS 2) IPS 3) SIEM 4) EDR	08
Q.2	Attempt any three.	
(a)	Write Short notes on Penetration testing life cycle.	08
(b)	Explain following Terms : 1) Dumpster Diving 2) War driving	08
(c)	What are the different password cracking techniques and solutions for hardening?	08
(d)	What do you mean by Secure code? Explain with one example.	08
Q.3	Attempt any three.	
(a)	Apply Polybius Square Cipher: 1) NFSU 2) BUS	08
(b)	Explain working of RSA algorithms.	08
(c)	Discuss working of AES algorithm.	08
(d)	Explain Elliptic Curve Cryptography.	08
Q.4	Attempt any two.	
(a)	Describe the four-way handshake process used in WPA2.	07
(b)	Explain 802.11 protocols.	07
(c)	You are tasked with securing a wireless network for a small business. Briefly outline the key steps you would take to implement a robust security solution, considering factors such as network size, user needs, and budget constraints.	07
Q.5	Attempt any two.	
(a)	Write short notes on OSCAR .	07

- (b) Discusses traffic acquisition Softwares and its importance for network forensic investigation.
- (c) Write a short Netflow..

07

07

--- End of Paper---

Seat No.: _____

Enrolment No. _____

NATIONAL FORENSIC SCIENCES UNIVERSITY

M.Sc. Cyber Security - Semester - II

End Semester Exam - July-2024

Subject Code: CTMSCS SII P2
Subject Name: Malware Analysis
Time: 2:30 PM - 5:30 PM

Date: 11/07/2024

Total Marks: 100

Instructions:

1. Write down each question on a separate page.
2. Attempt all questions.
3. Make suitable assumptions wherever necessary.
4. Figures to the right indicate full marks.

			Marks
Q.1		Attempt any three.	
	(a)	1. Explain function and property of obfuscated malware. 2. Define the function of PEiD and PEView tool in malware analysis.	08
	(b)	What are registers? Explain different types of registers with instruction example.	08
	(c)	Elaborate the Process Explorer and Process Monitor tools of sysinternal Utility.	08
	(d)	Explain following instructions with respect to X86: 1. mov eax, ebx 2. mov eax, 0x42 3. mov eax, [0x4037C4] 4. mov eax, [ebx] 5. mov eax, [ebx+esi*4] 6. cmp eax, 5 7. mov ecx, [x] 8. xor eax, 2	08
Q.2		Attempt any three.	
	(a)	Explain types of breakpoints: Software execution, Hardware execution, Conditional, and memory breakpoints.	08
	(b)	Diagnose what would be the C program from following assembly snapshot. 00401006 mov [ebp+var_4], 0 0040100D mov [ebp+var_8], 1 00401014 mov eax, [ebp+var_4] 00401017 add eax, 0Bh 0040101A mov [ebp+var_4], eax 0040101D mov ecx, [ebp+var_4] 00401020 sub ecx, [ebp+var_8]	08

		00401023 mov [ebp+var_4], ecx 00401026 mov edx, [ebp+var_4] 00401029 sub edx, 1 0040102C mov [ebp+var_4], edx 0040102F mov eax, [ebp+var_8] 00401032 add eax, 1 00401035 mov [ebp+var_8], eax 00401038 mov eax, [ebp+var_4] 0040103B cdq 0040103C mov ecx, 3 00401041 idiv ecx 00401043 mov [ebp+var_8], edx	
	(c)	(a) What is the goal of Malware analysis? (b) Describe the function of disassembler and debugger in malware analysis.	08
	(d)	Explain common executable DLLs of windows with function services.	08
Q.3		Attempt any three.	
	(a)	Discuss behaviour of hook injection while malware launching.	08
	(b)	Provide the practical explanation about setting up the malware analysis lab with required tool and technique with proper configurations.	08
	(c)	Differentiate between Static and Dynamic Analysis in detail.	08
	(d)	1. Differentiate between User mode and Kernel mode. 2. Compare Virus vs Worm.	08
Q.4		Attempt any two.	
	(a)	What is Cyber-Warfare? Explain with the help of a real-world case study with respect to Malwares.	07
	(b)	Explain PE File Headers and Sections.	07
	(c)	Explain malware sandboxing. How various parameters can be identified from sandbox-based malware analysis?	07
Q.5		Attempt any two.	
	(a)	Explain function of FLOSS and Regshot tool in Malware Analysis.	07
	(b)	Explain Stack memory with example.	07
	(c)	Write a note on reverse engineering tool and its effectiveness in malware analysis.	07

--- End of Paper---